# Alexandra J Weisberg


CONTACT

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PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, Oregon State University, Jeff Chang & Niklaus Grünwald labs

| | |
|---|---|
| Research Associate, Postdoc | 2019-Present |
| USDA NIFA Postdoctoral Fellow | 2017-2019 |
| Postdoctoral Scholar | 2015-2017 |

Research Assistant, Virginia Tech, Boris Vinatzer lab 2014

EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, VA, USA

Ph.D., Genetics, Bioinformatics, and Computational Biology 2010–2014

John Jelesko lab
Dissertation: Investigations into the molecular evolution of plant terpene, alkaloid, and urushiol biosynthetic enzymes assoc. with Translational Plant Sciences program

B.S., Computer Science 2006–2009

Mathematics minor

FIRST AUTHOR PUBLICATIONS

32 publications (8 first author)

Genomic Approaches to Plant-Pathogen Epidemiology and Diagnostics
Weisberg A. J., Grünwald N. J., Savory E. A., Putnam M. L., Chang J. H.
Annual Review of Phytopathology *59 (2021).*

Unexpected conservation and global transmission of agrobacterial virulence plasmids
Weisberg A. J., Davis E. W., Tabima J., Belcher M. S., Miller M., Kuo C.-H., Loper J. E., Grünwald N. J., Putnam M. L., Chang J. H.
Science *368.6495 (2020).*

A novel species-level group of *Streptomyces* exhibits variation in phytopathogenicity despite conservation of virulence loci
Weisberg A. J., Kramer C. G., Kotha R. R., Luthria D. L., Chang J. H., Clarke C. R.
Molecular Plant-Microbe Interactions *(2020) MPMI–06.*

Phytopathogenic *Rhodococcus* have diverse plasmids with few conserved virulence functions
Savory E. A.*, Weisberg A. J.*, Stevens D. M., Creason A. L., Fuller S. L., Pearce E. M., Chang J. H.
Frontiers in Microbiology *11 (2020) p. 1022.*

Sequencing and *de novo* assembly of the *Toxicodendron radicans* (poison ivy) transcriptome

Weisberg A. J., Kim G., Westwood J. H., Jelesko J. G.
Genes *8.11 (2017) p. 317.*

Evolutionary transitions between beneficial and phytopathogenic *Rhodococcus* challenge disease management

Savory E. A.*, Fuller S. L.*, Weisberg A. J.*, Thomas W. J., Gordon M. I., Stevens D. M., Creason A. L., Belcher M. S., Serdani M., Wiseman M. S., Grünwald N. J., Putnam M. L., Chang J. H.
eLife *6 (2017) e30925.*

Gall-ID: tools for genotyping gall-causing phytopathogenic bacteria

Davis II E. W.*, Weisberg A. J.*, Tabima J. F., Grünwald N. J., Chang J. H.
PeerJ *4 (2016) e2222.*

Similarity-based codes sequentially assigned to ebolavirus genomes are informative of species membership, associated outbreaks, and transmission chains

Weisberg A. J., Elmarakeby H. A., Heath L. S., Vinatzer B. A.
Open Forum Infectious Diseases *2 (2015) ofv024.*

** indicates co-first authorship*

## CO-AUTHOR PUBLICATIONS

Diversification of the Type VI Secretion System in Agrobacteria

Wu C.-F., Weisberg A. J., Davis E. W., Chou L., Khan S., Lai E.-M., Kuo C.-H., Chang J. H.
mBio *12.5 (2021) e01927–21.*

Experimental evolution can enhance benefits of rhizobia to novel legume hosts

Quides K. W., Weisberg A. J., Trinh J., Salaheldine F., Cardenas P., Lee H.-H., Jariwala R., Chang J. H., Sachs J. L.
Proceedings of the Royal Society B *288.1951 (2021) p. 20210812.*

Evolution of specialization in a plant-microbial mutualism is explained by the oscillation theory of speciation

Torres-Martínez L., Porter S. S., Wendlandt C., Purcell J., Ortiz-Barbosa G., Rothschild J., Lampe M., Warisha F., Le T., Weisberg A. J., Chang J. H., Sachs J. L.
Evolution *75.5 (2021) pp. 1070–1086.*

Recurrent mutualism breakdown events in a legume rhizobia metapopulation

Gano-Cohen K. A., Wendlandt C. E., Al Moussawi K., Stokes P. J., Quides K. W., Weisberg A. J., Chang J. H., Sachs J. L.
Proceedings of the Royal Society B *287.1919 (2020) p. 20192549.*

A deoR-type transcription regulator is required for sugar-induced expression of Type III secretion-encoding genes in *Pseudomonas syringae* pv. tomato DC3000

Turner S. E., Pang Y.-Y., O'Malley M. R., Weisberg A. J., Fraser V. N., Yan Q., Chang J. H., Anderson J. C.
Molecular Plant-Microbe Interactions *33.3 (2020) pp. 509–518.*

Antibacterial potential of secondary metabolites from Indonesian marine bacterial symbionts

Nofiani R., Weisberg A. J., Tsunoda T., Panjaitan R. G. P., Brilliantoro R., Chang J. H., Philmus B., Mahmud T.
International Journal of Microbiology *2020 (2020).*

Phenotypic characterization and phylogenetic analysis of *Pseudomonas syringae* strains associated with canker disease on apricot in Iran within the context of the global genetic diversity of the *P. syringae* complex

Vasebi Y., Khakvar R., Tian L., Moubarak P., Valentini F., Weisberg A. J., Vinatzer B. A.
European Journal of Plant Pathology *158 (2020) pp. 545–560.*

Biosynthesis of the nuclear factor of activated T cells inhibitor NFAT-133 in *Streptomyces pactum*

Zhou W., Posri P., Abugrain M. E., Weisberg A. J., Chang J. H., Mahmud T.
ACS Chemical Biology *(2020).*

The evolution, ecology, and mechanisms of infection by gram-positive, plant-associated bacteria

Thapa S. P., Davis E. W., Lyu Q., Weisberg A. J., Stevens D. M., Clarke C. R., Coaker G., Chang J. H.
Annual review of phytopathology *57 (2019) pp. 341–365.*

Re-evaluation of a Tn 5:: gacA mutant of *Pseudomonas syringae* pv. tomato DC3000 uncovers roles for uvrC and anmK in promoting virulence

O'Malley M. R., Weisberg A. J., Chang J. H., Anderson J. C.
PLOS One *14.10 (2019) e0223637.*

Genomic and metabolic differences between *Pseudomonas putida* populations inhabiting sugarcane rhizosphere or bulk soil

Lopes L. D., Weisberg A. J., Davis E. W., Varize C. d. S., Pereira e Silva M. d. C., Chang J. H., Loper J. E., Andreote F. D.
PLOS One *14.10 (2019) e0223269.*

Tropical soils are a reservoir for fluorescent *Pseudomonas* spp. biodiversity

Lopes L. D., Davis E. W., Pereira e Silva M. d. C., Weisberg A. J., Bresciani L., Chang J. H., Loper J. E., Andreote F. D.
Environmental microbiology *20.1 (2018) pp. 62–74.*

Response to comments on "Evolutionary transitions between beneficial and phytopathogenic *Rhodococcus* challenge disease management"

Chang J. H., Putnam M. L., Grünwald N. J., Savory E. A., Fuller S. L., Weisberg A. J..
eLife *7 (2018) e35852.*

Transient heterologous gene expression methods for poison ivy leaf and cotyledon tissues

Dickinson C. C., Weisberg A. J., Jelesko J. G.
HortScience *53.2 (2018) pp. 242–246.*

The population genetic test Tajima's D identifies genes encoding pathogen-associated molecular patterns and other virulence-related genes in *Ralstonia solanacearum*

Eckshtain-Levi N., Weisberg A. J., Vinatzer B. A.
Molecular plant pathology *19.9 (2018) pp. 2187–2192.*

Genome variations between rhizosphere and bulk soil ecotypes of a *Pseudomonas koreensis* population

Lopes L. D., Pereira e Silva M. d. C., Weisberg A. J., Davis E. W., Yan Q., Varize C. d. S., Wu C.-F., Chang J. H., Loper J. E., Andreote F. D.
Environmental microbiology *20.12 (2018) pp. 4401–4414.*

Evolution of the US biological select agent *Rathayibacter toxicus*

Davis E. W., Tabima J. F., Weisberg A. J., Lopes L. D., Wiseman M. S., Wiseman M. S., Pupko T., Belcher M. S., Sechler A. J., Tancos M. A., Schroeder B. K., Murray T. D., Luster D. G., Schneider W. L., Rogers E. E., Andreote F. D., Grünwald N. J., Putnam M. L., Chang J. H.
MBio *9.4 (2018).*

Association mapping, transcriptomics, and transient expression identify candidate genes mediating plant-pathogen interactions in a tree

Muchero W., Sondreli K. L., Chen J.-G., Urbanowicz B. R., Zhang J., Singan V., Yang Y., Brueggeman R. S., Franco-Coronado J., Abraham N., Yang J.-Y., Moremen K. W., Weisberg A. J., Chang J. H., Lindquist E., Barry K., Ranjan P., Jawdy S., Schmutz J., Tuskan G. A., LeBoldus J. M.
Proceedings of the National Academy of Sciences *115.45 (2018) pp. 11573–11578.*

A proposal for a genome similarity-based taxonomy for plant-pathogenic bacteria that is sufficiently precise to reflect phylogeny, host range, and outbreak affiliation applied to *Pseudomonas syringae* sensu lato as a proof of concept

Vinatzer B. A., Weisberg A. J., Monteil C. L., Elmarakeby H. A., Sheppard S. K., Heath L. S.
Phytopathology *107.1 (2017) pp. 18–28.*

Morphological and transcriptomic evidence for ammonium induction of sexual reproduction in *Thalassiosira pseudonana* and other centric diatoms

Moore E. R., Bullington B. S., Weisberg A. J., Jiang Y., Chang J., Halsey K. H.
PLOS One *12.7 (2017) e0181098.*

Isothermal amplification and lateral-flow assay for detecting crown-gall-causing *Agrobacterium* spp.

Fuller S. L., Savory E. A., Weisberg A. J., Buser J. Z., Gordon M. I., Putnam M. L., Chang J. H.
Phytopathology *107.9 (2017) pp. 1062–1068.*

Microbe-ID: an open source toolbox for microbial genotyping and species identification

Tabima J. F., Everhart S. E., Larsen M. M., Weisberg A. J., Kamvar Z. N., Tancos M. A., Smart C. D., Chang J. H., Grünwald N. J.
PeerJ *4 (2016) e2279.*

Genome-enabled phylogeographic investigation of the quarantine pathogen *Ralstonia solanacearum* race 3 biovar 2 and screening for sources of resistance against its core effectors

Clarke C. R., Studholme D. J., Hayes B., Runde B., Weisberg A., Cai R., Wroblewski T., Daunay M.-C., Wicker E., Castillo J. A., Vinatzer B. A.
Phytopathology *105.5 (2015) pp. 597–607.*

Phylobiochemical characterization of class-Ib aspartate/prephenate aminotransferases reveals evolution of the plant arogenate phenylalanine pathway

Dornfeld C., Weisberg A. J., KC R., Dudareva N., Jelesko J. G., Maeda H. A.
Plant Cell *26.7 (2014) pp. 3101–3114.*

MANUSCRIPTS IN PREP OR UNDER REVISION

"Diversification of plasmids in a genus of pathogenic and nitrogen-fixing bacteria" Weisberg A.J., Miller M., Ream W., Grunwald N.J., and Chang J.H. in press. Philosophical Transactions B special issue "The secret lives of microbial mobile genetic elements."

”Pangenome evolution reconciles robustness and instability of symbioses” Weisberg A.J., Rahman A., Backus D., Tyavanagimatt P., Chang J.H., Sachs J.L. manuscript in prep.

”Dynamic interactions between mega-sized symbiosis integrative and conjugative elements and bacterial chromosomes” Weisberg A.J., Sachs J.L., Chang J.H. manuscript in prep.

”*Streptomyces canutascabiei* sp. nov., which causes potato common scab and is distributed across the world” Clarke C., Nguyen H.P., Weisberg A.J., Chang J.H. submitted to IJSEM.

”Dual Adhesive Unipolar Polysaccharides Synthesized by Overlapping Biosynthetic Pathways in *Agrobacterium tumefaciens*” Onyeziri M.C., Natarajan R., Hardy G.G., Xu J., Reynolds I.P., Kim J., Merritt P.M., Danhorn T., Hibbing M.E., Weisberg A.J., Chang J.H., Fuqua C. Preprint available at https://www.biorxiv.org/content/10.1101/2021.04.22.440995v1

”Modular evolution of secretion systems and virulence plasmids in a bacterial species complex” Chou L., Lin Y.C., Haryono M., Santos M.N.M., Cho S.T., Weisberg A.J., Wu C.F., Chang J.H., Lai E.M, Kuo C.H. Preprint available at https://www.biorxiv.org/content/10.1101/2021.05.20.444927v2

## FUNDING & FELLOWSHIPS

| | | |
|---|---|---|
| USDA NIFA Postdoctoral Fellowship | $152,000 | 2016 |
| MPS program mini grant for dept. purchase of high memory server for assembly and transcriptomics. Kim G, Weisberg A, Fedkenheuer M, Fedkenheuer K, Fang Y | $24,000 | 2013 |
| Molecular Plant Sciences program travel award for Workshop on Molecular Evolution at Woods Hole | $2,000 | 2011 |
| Virginia Tech Scholars Award | $2,000 | 2006 |

## TEACHING

**Instructor of record**, BI 311 Genetics — Winter 2020

- Required for undergraduates majoring in the life sciences
  173 students
  4 credit lecture and recitation

**Nanopore sequencing workshop**, Oregon State University — Dec 2019
*Led a workshop and discussion group on nanopore sequencing, including DNA extraction techniques, experiences with MinION sequencing, and ease of use.*

**Microbial comparative genomics and Publication quality figures in R workshops**, Virginia Tech Nov 2019
*Invited to teach two workshops, one on microbial comparative genomics techniques and analyses, and another on making publication-quality figures in R. Graduate students in the Translational Plant Sciences program attended and performed analyses with example data.*

**Guest lecture, Microbial Forensics and Biosecurity**, Virginia Tech Nov 2019
*Invited as guest lecturer on Agrobacterium and agricultural epidemiology methods and challenges for an undergraduate PPWS 4114 Microbial Forensics and Biosecurity class at Virginia Tech.*

**Microbiome Training Workshop**, Oregon State University May 2017
*TA for a workshop on analyzing microbiome data.*

## OUTREACH

**DNA Biology and Bioinformatics STEM Camp** Jul 2019
David Douglas High School/Nestucca High School, Oregon State University
*Developed and led lab and bioinformatics sections as part of a 3 day summer STEM camp for underrepresented high school students from Portland, OR and Cloverdale, OR. Students learned to transcribe DNA to RNA and translate protein sequences, identify genes with BLAST, and examine 3D protein structure.*

**ASE Midsummer Conference Workshop** Oregon State University Aug 2018
*Taught as part of a hands-on workshop for high school STEM interns on the history of agriculture titled "AnthroBiology, Mutants, GMOs, and Big Data - The History of Food", examining the relationship of corn to teosinte and modern genetics of sweet corn.*

**DNA Biology and Bioinformatics STEM Camp** Jul 2018
David Douglas High School, Oregon State University
*Developed and led lab and bioinformatics sections as part of a 3 day summer STEM camp for underrepresented high school students from Portland, OR. Students learned to transcribe DNA to RNA and translate protein sequences, identify genes with BLAST, and examine 3D protein structure.*

**DNA Biology and Bioinformatics STEM camps**, Oregon State University 2015-2017
*Developed and led lab sections and taught bioinformatics classes as part of 2-week summer STEM camps targeted at local and Portland,OR high school students.*
*- Students learned to extract DNA, run PCR, and perform either genome sequencing and assembly (2015,2016) or microbiome sequencing and analysis (2017).*
*- Camps were developed to particularly encourage women and other underrepresented groups to consider pursuing a career in STEM fields.*

## MENTORING

**Undergraduates and Postbacc.**
Riely White, Elizabeth Riutta, Emma Pearce, Kaavya Subramanian, Casey Huggins, Joey Edgerton, Dakota Backus, Heidi Lederhos, Michael Belcher, Ash Gill - *Oregon State University*
Oscar Tanner - *Virginia Tech*
**High School students**
Ella Rose, Parinita Tyavanagimatt

**Visiting graduate students and postdocs**
Kenjiro Quides, Lucas Dantas, Lorena Torres Martinez, Angela Porter

## PROFESSIONAL SERVICE

Reviewer for the journals:
- Molecular Biology and Evolution (MBE)
- Molecular Plant-Microbe Interactions (MPMI)
- mSystems
- New Phytologist
- Phytopathology
- Science Advances

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| Postdoc member of departmental seminar committee | 2021 |
| Assisted with managing departmental computing infrastructure | 2015-Present |
| Postdoc member of search committee for tenure track faculty position in the Department of Botany and Plant Pathology | 2018 |
| Plant Pathology, Physiology, and Weed Science GSO Secretary/Treasurer, Virginia Tech | 2012-2014 |
| Co-coordinator for Molecular Plant Science Discussion Group, Virginia Tech | 2012-2014 |
| Molecular Plant Sciences GSO Bioinformatics Committee member, Virginia Tech | 2012-2014 |
| Co-organized Workshop on whole genome/RNA-seq, Virginia Tech | Apr 2013 |
| Co-organized PPWS Research Symposium, Virginia Tech | Oct 2012 |

## MEMBERSHIPS

IS-MPMI - International Society for Molecular Plant-Microbe Interactions

## INVITED TALKS

**Mid-Atlantic Plant Molecular Biology Society 2021** Towson, VA — Aug 2021
*MAPMBS 2021 meeting invited talk "Modularity underpins the evolution of nitrogen fixing symbiosis"*

Phytobiomes Alliance webinar — Jul 2020
*"Genomic approaches to the epidemiology and diagnostics of phytopathogens: tracking the global spread of agrobacteria"*
*Viewed by 75 attendees, >100 views online.*
*https://www.youtube.com/watch?v=7d6oKrtXBsU*

**Departmental Seminar** Virginia Tech, Blacksburg, VA — Nov 2019
*Genetics, Bioinformatics, and Computational Biology department seminar "Evolution and epidemiology of the natural genetic engineer Agrobacterium"*

**Departmental Seminar** University of Arizona, Tucson, AZ — Sep 2018
*School of Plant Sciences Departmental Seminar "Evolution of the Natural Genetic Engineer: Agrobacterium"*

## CONFERENCES & PRESENTATIONS

**Departmental Seminar** Oregon State University, Corvallis, OR — Jan 2021
*Botany and Plant Pathology Departmental Seminar "Modularity Underpins the Evolution of Nitrogen Fixing Symbiosis"*

**Bioinformatics Users Group (BUG)** Oregon State University, Corvallis, OR May 2020
*"Getting started with Nanopore sequencing in the lab: i.e. you can do it, and it's not a random sequence generator anymore!"*

**Departmental Seminar** Oregon State University, Corvallis, OR Apr 2020
*Botany and Plant Pathology Departmental Seminar "Mobilome flexibility confers robustness to good and bad symbionts of plants" tag team with Jeff Chang*

**39th Crown Gall Conference** University of Missouri, Columbia, MO Oct 2019
*"Evolution and epidemiology of the natural genetic engineer Agrobacterium"*

**CGRB Fall Conference** Oregon State University, Corvallis, OR Sep 2019
*lightning talk "Evolution and epidemiology of the natural genetic engineer Agrobacterium"*

**2019 IS-MPMI XVIII Congress** Glasgow, Scotland Jul 2019
*poster "The genomic bases for cheating among wild populations of nodule-associated Bradyrhizobium"*

**2019 IS-MPMI XVIII Congress** Glasgow, Scotland Jul 2019
*satellite conference invited talk/workshop "Genome Visualization Tools"*

**9th Yosemite Symbiosis Workshop** Yosemite, CA May 2019
*"The genomic bases for cheating among wild populations of nodule-associated Bradyrhizobium"*

**Plant & Animal Genome Conference XXVII** San Diego, CA Jan 2019
*"Evolution of the Natural Genetic Engineer: Agrobacterium"*

**NIFA Fellows Project Director's Meeting** USDA NIFA, Washington, DC Aug 2018
*poster "Characterizing Competitiveness And Persistence Of Agrobacteria In The Crown Gall Disease Niche"*

**Bioinformatics Users Group (BUG)** Oregon State University, Corvallis, OR Nov 2017
*"Characterizing the population genetic history of Rhodococcus and tracking disease outbreaks in the nursery industry"*

**38th Crown Gall Conference** Oregon State University, Corvallis, OR Oct 2017
*"Evolution of the Agrobacterium secondary and tertiary replicons"*

**Departmental Seminar** Oregon State University, Corvallis, OR Mar 2017
*Botany and Plant Pathology Departmental Seminar "Modeling the evolution of Agrobacterium Ti plasmids"*

**37th Crown Gall Conference** Indiana University, Bloomington, IN Oct 2016

*"The characterization of novel Ti plasmid types"*

**CGRB Fall Conference** Oregon State University, Corvallis, OR — Sep 2016
*poster "Ti plasmids cluster genetically and cluster by Agrobacterium taxa and plant host type"*

**2016 IS-MPMI XVII Congress** Portland, OR — Jun 2016
*poster "Ti plasmids cluster genetically and cluster by Agrobacterium taxa and plant host type"*

**36th Crown Gall Conference** DowAgrosciences, Zionsville, IN — Nov 2015
*"Ti plasmids cluster according to opine type and are augmented by a large gene pool"*

**CGRB Fall Conference** Oregon State University, Corvallis, OR — Sep 2015
*poster "Deep sequencing of crown gall-forming Agrobacterium genera reveals diversity in population structure and Ti plasmid sequence"*

**MPS Mini-Symposium** Virginia Tech, Blacksburg, VA — Feb 2014
*"De novo assembly of the Toxicodendron radicans (poison ivy) transcriptome"*

**Departmental seminar** Virginia Tech, Blacksburg, VA — Nov 2013
*"De novo assembly of the Toxicodendron radicans (poison ivy) transcriptome"*

**Departmental seminar** Virginia Tech, Blacksburg, VA — Apr 2013
*"Examining the molecular evolution of neofunctionalization in plant secondary metabolism"*

**MPS Mini-Symposium** Virginia Tech, Blacksburg, VA — Mar 2012
*poster "Identifying effectively-mutable protein space using measures of natural selection: a meta-analysis of two terpene synthase gene families"*

**Banff Conference on Plant Metabolism** Banff, Alberta, Canada — Jun 2010
*poster "Molecular evolution of plant tyrosine and phenylalanine biosynthesis"*

**MPS Mini-Symposium** Virginia Tech, Blacksburg, VA — Jan 2010
*poster "Molecular evolution of plant tyrosine and phenylalanine biosynthesis"*

SOCIAL MEDIA & PRESS

For *"Unexpected conservation and global transmission of agrobacterial virulence plasmids"* Science 368, 6495, eaba5256:
https://www.sciencedaily.com/releases/2020/06/200604152111.htm
https://www.eurekalert.org/pub_releases/2020-06/udoa-sdt052720.php
https://www.eurekalert.org/pub_releases/2020-06/osu-pdd052920.php
https://www.agdaily.com/crops/researchers-develop-new-way-trace-global-spread-major-plant-disease/
https://phys.org/news/2020-06-global-major-disease.html
https://www.seedquest.com/solutions.php?type=solutionid_article=117914
https://www.technology.org/2020/06/05/successful-detailed-tracking-of-major-plant-diseases-global-spread/
https://www.nurserymag.com/article/scientitsts-successfully-track-detailed-spread-of-agrobacterium/
https://www.courthousenews.com/plant-detectives-develop-tool-to-track-spread-of-major-crop-disease/
https://www.producegrower.com/article/scientitsts-successfully-track-detailed-spread-of-agrobacterium/
https://www.farmprogress.com/crops/stopping-plant-disease

For *"Evolutionary transitions between beneficial and phytopathogenic Rhodococcus challenge disease management"* eLife 6:e30925:
Commentary in:
Melnyk, RA and Haney, CH (2017) Plasmid-powered evolutionary transitions. eLife 6:e33383.
Tena, G (2018) Dr. Rhodo and Mr. Coccus. Nature Plants. 4, 8(2018)

Podcast in Bacteriofiles:
https://www.asm.org/index.php/podcasts/bacteriofiles/item/7305-plasmid-promotes-plant-pathogenesis-bacteriofiles-340

Press releases in:
http://www.nurserymag.com/article/bacterium-breakthrough/
https://content.govdelivery.com/accounts/USDANIFA/bulletins/1d37681
https://blog.plantwise.org/2017/12/15/rhodococcus-jekyll-and-hyde-bacteria-discovered-in-plant-nurseries/
http://extension.oregonstate.edu/news/release/2017/12/osu-plant-pathologists-discover-unusual-evolutionary-transition-common-bacteria
http://www.nurserymag.com/article/osu-discover-rhodococcus-bacteria-evolves/
https://www.sciencedaily.com/releases/2017/12/171212125356.htm
http://www.capitalpress.com/Nursery/20171219/researchers-developing-tools-to-track-harmful-bacteria
https://phys.org/news/2017-12-pathologists-unusual-evolutionary-transition-common.html
https://plantae.org/evolutionary-transitions-between-beneficial-and-phytopathogenic-rhodococcus/
http://www.westernfarmerstockman.com/technology/researchers-uncover-bacteria-change-tactic

Interview about computing infrastructure and research at Oregon State featured in AMD keynote address at CES 2019

REFERENCES

**Jeff H. Chang PhD**
Postdoctoral co-advisor (lead).
Professor, Botany and Plant Pathology. Oregon State University
541.737.5278
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4575 SW Research Way, Corvallis OR, 97333

**Niklaus Grünwald PhD**
Postdoctoral co-advisor (secondary).
Research Plant Pathologist, USDA ARS
541.738.4049
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**John G. Jelesko PhD**
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**Boris Vinatzer PhD**
Mentor.
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